Exponential online mistakes for isotropic Bayesian networks with a monotone Lipschitz activation

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Abstract

We give a fixed strictly increasing, 1-Lipschitz activation for which an isotropic Gaussian prior on network weights makes superpolynomially many online mistakes on a fixed input permutation. The activation is $\psi(t) = (2t + \phi(t))/3$, where ϕ is the period-four triangle wave; its slopes are $1/3$ and 1. For every $m \geq 5$, the target has $d = \Theta(m^6)$ Boolean inputs, a bounded-fan-in Boolean circuit of size $\Theta(d)$, and an explicit one-hidden-layer teacher of width $8m - 4$ with every weight at most one and output margin one. At every sufficiently large learner width, taking the Gaussian variance to zero at that fixed width makes the actual posterior probability of the correct label tend to zero at each of 2^{m-4} designated rounds. The learner conditions on absolute output margin greater than $1/2$ on the full Boolean cube before conditioning on exact labels. The proof identifies an exact weighted Fourier variation norm, excludes competing sign functions outside an explicitly defined parity family by a calibration and margin argument that controls the affine part of every competing network, and strictly orders the remaining alternatives, each wrong on one logical input class, using distinct square roots. Integer neuron allocations transfer the order to every sufficiently large finite width, and Gaussian energy separation controls the posterior mixture. The mistake bound is exponential in the displayed teacher width and is $\exp(\Omega(d^{1/6}))$ in input dimension and Boolean-circuit size. Calibration occupies 15/16 of the cube, and the sufficient width is not claimed to be polynomial. The theorem does not establish the corresponding statement for tanh, ReLU, GELU, or hardtanh.

1 Model, target, and theorem

In realizable online prediction, an input arrives, the learner predicts its label, and then the true label is revealed. A polynomial total-mistake guarantee would bound the number of errors in terms of target complexity, independently of the input order and the sequence length. Our sequence has a long calibration prefix, but the designated errors after that prefix alone exceed every polynomial in the target's circuit size. The question is different from IID sample complexity, where the training inputs are drawn randomly from a fixed distribution.

Let ϕ be the odd triangle wave of period four, defined by

$$\phi(t) = t \quad (-1 \leq t \leq 1), \qquad \phi(t+2) = -\phi(t).$$

The learner uses the fixed activation

$$\psi(t) = \frac{2t + \phi(t)}{3} = \alpha t + \varepsilon\phi(t), \qquad \alpha = \frac{2}{3}, \quad \varepsilon = \frac{1}{3}. \tag{1}$$

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It is odd, continuous, strictly increasing, and 1-Lipschitz, with slopes $1/3$ and 1. It is unbounded and vanishes at zero. This same activation is used at every value of the construction parameter. Triangle-wave activations have previously been studied in Bayesian neural networks by Meronen, Trapp, and Solin [1, Section 3.2]. Our triangle normalization is their equation (11) with period four. The affine-plus-triangle activation (1) is specified explicitly here; we do not assert that this monotone combination is a standard activation from that work, and no theorem about its prior is imported from that reference. On $X = \{-1, 1\}^d$, write $\bar{x} = (1, x)$ and use one hidden layer,

$$F_{W,v}(x) = \sum_{a=1}^{N} v_a \psi(W_a \cdot \bar{x}), \qquad W_{ai} \sim N(0, N\sigma^2/d), \quad v_a \sim N(0, \sigma^2/N), \tag{2}$$

independently, including the bias coordinate $i = 0$. There is no coordinate-dependent covariance. Equivalently, the whitened coordinates $\xi = \sqrt{d/N}\, W$ and $\eta = \sqrt{N}\, v$ are independent centered Gaussians with common variance σ^2. The Gaussian energy is

$$K_N(W, v) = \frac{d}{N}\|W\|_F^2 + N\|v\|_2^2. \tag{3}$$

Before observing labels the learner conditions on

$$\mathcal{M} = \{|F_{W,v}(x)| > 1/2 \text{ for every } x \in X\}. \tag{4}$$

It subsequently conditions on exact noiseless training signs. All posterior statements below refer to this actual conditional measure, not to a selected minimum-energy network. The bounded-weight teacher complexity $C_2(f)$ is the least integer $C \geq 1$ for which a width-C network in the same unnormalized predictor form as (2), with all incoming and readout weights at most C in absolute value, implements f with margin greater than $1/2$.

Fix $m \geq 5$, and set

$$q_i = 2m + 2i - 1, \quad r_i = 2q_i \quad (1 \leq i \leq m), \quad D = \sum_i r_i = 6m^2, \quad c = 10^8(D+1)^3 + 1, \quad d = c + D. \tag{5}$$

Partition the coordinates into a common core C of odd size c and blocks B_i of even size r_i. For a set A, write $\chi_A(x) = \prod_{j \in A} x_j$. Define $z(x) \in G := \mathbb{F}_2^m$ by

$$(-1)^{z_i(x)} = \chi_{B_i}(x).$$

Our target is

$$f_m(x) = \chi_C(x)\big(2\mathbf{1}_{\{z(x)=0\}} - 1\big). \tag{6}$$

Theorem 1. *For every $m \geq 5$, the target (6) and a fixed permutation of the full cube have the following properties.*

1. *The target has a bounded-fan-in Boolean circuit of size $\Theta(d)$ and an explicit one-hidden-layer ψ teacher of width $8m - 4$, with all incoming and readout weights bounded by one and absolute output margin at least one. In particular, $C_2(f_m) \leq 8m - 4$.*

2. *The permutation has a calibration prefix of length $15 \cdot 2^{d-4}$, followed by 2^{m-4} designated, nonrepeated inputs. There is $N_0(m)$ such that for every fixed $N \geq N_0(m)$ and every designated round,*

$$\lim_{\sigma \downarrow 0} \mathbb{P}_{N,\sigma}\big(F(x) f_m(x) > 0 \mid \mathcal{M}, \textit{all preceding exact labels}\big) = 0. \tag{7}$$

The permutation is independent of N and σ.

3. *At each such fixed width, posterior-label majority makes all 2^{m-4} designated mistakes for all sufficiently small σ; the posterior wrong-label probability also exceeds $2/3$ there. The probability that posterior sampling makes every designated mistake tends to one.*

The number of designated mistakes is exponential in the displayed teacher width, and is $\exp(\Omega(d^{1/6}))$. *The theorem asserts no variance threshold uniform in width and no polynomial bound on $N_0(m)$.*

Corollary 2. *For the explicit unit-bounded teacher widths $w_m = 8m - 4$, the construction forces $2^{(w_m-28)/8}$ designated mistakes, at every sufficiently large learner width in the fixed-width strong-prior limit. Its input dimension and bounded-fan-in Boolean-circuit size are $\Theta(w_m^6)$.*

The calibration and width qualifications are substantive. The result rules out an arbitrary-order polynomial mistake bound in the target's Boolean-circuit size, but does not give $2^{\Omega(d)}$ mistakes or a short calibration prefix. All proofs are conventional mathematical proofs; no formal verification is claimed.

2 An exact weighted Fourier norm

For a cube function F, let $\widehat{F}(A) = \mathbb{E}F(X)\chi_A(X)$. For $s \in G$, set

$$A_s = C \cup \bigcup_{i:s_i=1} B_i, \qquad a_s = \sqrt{|A_s|} = \sqrt{c + \sum_i r_i s_i}.$$

The code projection is the orthogonal projection onto these characters:

$$PF(x) = \sum_{s\in G} \widehat{F}(A_s)\chi_{A_s}(x) = \chi_C(x)\,\bar{F}(z(x)). \tag{8}$$

Equivalently, $\bar{F}(z) = \mathbb{E}[\chi_C(X)F(X) \mid z(X) = z]$. Define

$$Q(F) = \sum_{s\in G} a_s|\widehat{F}(A_s)|, \qquad \|F\|_{\mathcal{V}_\rho} = \inf\left\{\sum_a |v_a|\|\theta_a\|_2 : F = \sum_a v_a\rho(\theta_a\cdot\bar{x})\right\}, \quad \rho \in \{\phi,\psi\},$$

where representations are finite and equality is on the cube. We abbreviate $\|F\|_{\mathcal{V}} = \|F\|_{\mathcal{V}_\psi}$ for the learner's variation cost. The triangle cost is finite on every cube function: odd-support characters use bias zero, even-support characters use bias one, and the constant uses $\phi(1) = 1$. The analogous finiteness for ψ follows from the orientation construction below; constants and degree-one characters use $\psi(1) = 1$ and $\psi(x_i) = x_i$ directly. Homogeneity and the triangle inequality follow by scaling and concatenating representations. For either activation, $\|F\|_\infty \le \sqrt{d+1}\|F\|_{\mathcal{V}_\rho}$ proves positivity.

Theorem 3 (Atomic norm and quantitative rounding)**.** *More generally, let $c \ge 1000$ be odd, let the disjoint blocks have even sizes at least four and total size D, and assume $c \ge 100(D+1)^2$. For every biased triangle-wave ridge $h_\theta(x) = \phi(\theta\cdot\bar{x})$,*

$$Q(h_\theta) \le \|\theta\|_2. \tag{9}$$

Nonzero equality occurs exactly at the minimal pure-character atoms: zero bias, weights ±1 on the core and on a union of whole blocks, and zero weights elsewhere. Consequently,

$$\left\|\sum_{s\in G} b_s\chi_{A_s}\right\|_{\mathcal{V}_\phi} = \sum_{s\in G} a_s|b_s|. \tag{10}$$

Moreover, with $\kappa = 1/200$, every atom has a number $e_\theta \geq 0$ such that

$$\|\theta\|_2 - Q(h_\theta) \geq \kappa e_\theta, \tag{11}$$

and the following two bounds hold:

$$|h_\theta(x) - h_\theta(x')| \leq e_\theta \|x - x'\|_2 \qquad \text{if } (\chi_C, z)(x) = (\chi_C, z)(x'), \tag{12}$$

$$|\mathbb{E}_{X_C} h_\theta(X_C, u)| \leq \sqrt{D+1}\, e_\theta \qquad \text{for every fixed physical block assignment } u. \tag{13}$$

Also, with the uniform cube L^2 norm,

$$\|(I - P)h_\theta\|_2 \leq e_\theta. \tag{14}$$

The exactness and the quantitative statement are both needed. An approximate norm identity need not preserve the small strict energy differences used later. A proof, including constants, is given in Appendix A.

For odd $|A|$, the pure atom is especially simple:

$$\phi\left(\sum_{i \in A} x_i\right) = (-1)^{(|A|-1)/2} \chi_A(x). \tag{15}$$

Thus all witnesses in (10) use ordinary bias-free triangle-wave neurons with incoming weights zero or one, with a sign absorbed into the readout.

Lemma 4 (Four orientations cancel the affine part)**.** *For every odd support A of size at least three, there are four vectors $u_1, \ldots, u_4$, supported on A with nonzero entries ± 1, such that*

$$\frac{1}{4} \sum_{j=1}^{4} \psi(u_j \cdot x) = \varepsilon (-1)^{(|A|-1)/2} \chi_A(x).$$

Consequently the learner's exact code norm is

$$\left\| \sum_s b_s \chi_{A_s} \right\|_{\mathcal{V}} = \frac{1}{\varepsilon} \sum_s a_s |b_s|. \tag{16}$$

Proof. On three coordinates choose the four sign rows $(+++), (+--), (-+-), (--+)$. Every row product is positive and every column sum is zero. Pair the remaining coordinates, assigning the same balanced column $(++--)$ to both coordinates of each pair. The row products remain positive and all column sums remain zero. Thus the four affine terms cancel, while all four triangle ridges equal the same signed parity by (15). A character with coefficient b_s is represented using four readouts of absolute value $|b_s|/(4\varepsilon)$ and incoming norms a_s.

For the lower bound, every code character has degree at least three, so affine functions have zero code projection. Thus $Q(\psi(\theta \cdot \bar{x})) = \varepsilon Q(\phi(\theta \cdot \bar{x})) \leq \varepsilon \|\theta\|_2$. Summing over neurons proves the matching lower bound. The same orientation construction with the bias coordinate as one of the odd support positions represents even-support physical characters, as used in the norm-finiteness observation above. □

3 Calibration controls the affine component

The code projection alone does not preserve global absolute margin. Moreover, an individual ψ ridge contains an affine term that need not be small when its triangle component is close to a pure character. The calibration below controls that entire affine component of the network without assuming cancellation in a competitor.

Let $M = 2^m$, and define

$$H = \{z : z_1 = z_2 = z_3 = z_4 = 0\}, \qquad U = e_1 + H, \qquad L_H = H \cup U. \tag{17}$$

Thus $|U| = M/16$ and $0 \notin U$. The calibration reveals the target on every physical input with $z(x) \notin U$, exactly $15 \cdot 2^{d-4}$ inputs. Write $\mathcal{K} = G \setminus U$ for the known logical inputs, and $y(z) = 2\mathbf{1}_{\{0\}}(z) - 1$.

Lemma 5 (The flat calibration bound)**.** *Every real $f : G \to \mathbb{R}$ with $f(0) \geq 1/2$ and $f(z) \leq -1/2$ for $z \notin \{0\} \cup U$ satisfies*

$$\sum_s |\widehat{f}(s)| \geq L_0 := \frac{3}{2} - \frac{4}{M}. \tag{18}$$

For every $v \in U$, the star score

$$f_v = -\frac{1}{2} + \mathbf{1}_{\{0\}} + \mathbf{1}_{\{v\}} \tag{19}$$

attains equality and has global margin $1/2$.

Proof. Use the dual function

$$\lambda(z) = M\mathbf{1}_{\{0\}}(z) - 2 - 2\mathbf{1}_H(z) + 2\mathbf{1}_U(z), \qquad \mathcal{D}(f) = \mathbb{E}_z \lambda(z) f(z).$$

Its values are $M - 4$ at zero, -4 on $H \setminus \{0\}$, zero on U, and -2 elsewhere. Its Fourier multipliers are -1 at zero, $+1$ on $L_H^\perp \setminus \{0\}$, $3/4$ on $H^\perp \setminus L_H^\perp$, and $+1$ outside $H^\perp$. Thus $\mathcal{D}(f) \leq \sum_s |\widehat{f}(s)|$, while the training inequalities give $\mathcal{D}(f) \geq \frac{1}{2}\mathbb{E}|\lambda| = L_0$. Finally,

$$\widehat{f}_v(0) = -\frac{1}{2} + \frac{2}{M}, \qquad \widehat{f}_v(s) = \frac{1 + (-1)^{s \cdot v}}{M} \quad (s \neq 0),$$

whose absolute sum is L_0. □

Lemma 6 (Calibrated non-code gap)**.** *Suppose F has global margin at least $1/2$ and agrees with the calibration. If its sign is not of the form*

$$\operatorname{sign} F(x) = \chi_C(x) g(z(x)), \qquad g : G \to \{-1, 1\}, \tag{20}$$

then

$$\varepsilon \|F\|_{\mathcal{V}} \geq \sqrt{c}\, L_0 + \frac{\kappa}{10\sqrt{D+1}}. \tag{21}$$

Proof. Fix any finite representation $F = \sum_a v_a \psi(\theta_a \cdot \bar{x})$ and write

$$F = L + \varepsilon T, \qquad L(x) = \alpha \sum_a v_a \theta_a \cdot \bar{x}, \qquad T = \sum_a v_a \phi(\theta_a \cdot \bar{x}).$$

Let A be the Euclidean norm of the affine coefficient vector of L, put $V = \sum_a |v_a| \|\theta_a\|_2$, $B = \sum_a |v_a| e_{\theta_a}$, and $R = T - PT$. The triangle atomic estimates give

$$V \geq \frac{Q(F)}{\varepsilon} + \kappa B, \qquad \|R\|_2 \leq B. \tag{22}$$

Indeed $PL = 0$, $Q(F) = \varepsilon Q(T)$, and the Fourier triangle inequality gives the first assertion; (14) gives the second.

Write $PF = \chi_C \bar{F}(z)$. Calibration on complete physical fibers implies $y(z)\bar{F}(z) \geq 1/2$ on $\mathcal{K}$. Put $s(z) = y(z)\bar{F}(z) - 1/2 \geq 0$ there. On a known logical fiber the random variable $Y = \chi_C(X)[L(X) + \varepsilon R(X)]$ has conditional mean zero, because $P(L + \varepsilon R) = 0$. The training margin gives $s(z) \geq -y(z)Y$ pointwise. Hence

$$s(z) \geq \frac{1}{2}\mathbb{E}[|L + \varepsilon R| \mid z(X) = z].$$

Since $|\lambda| \geq 2$ on $\mathcal{K}$, the flat dual satisfies

$$\mathcal{D}(\bar{F}) - L_0 = \mathbb{E}_z \mathbf{1}_{\mathcal{K}}(z)|\lambda(z)|s(z) \geq \mathbb{E}\big[\mathbf{1}_{\{z(X)\in\mathcal{K}\}}|L + \varepsilon R|\big]. \tag{23}$$

All expectations in the last expression are unconditional uniform-cube expectations, with the displayed indicator.

For an affine Rademacher sum of coefficient norm A, $\mathbb{E}L^2 = A^2$ and $\mathbb{E}L^4 \leq 3A^4$. Moment interpolation yields $\mathbb{E}|L| \geq A/\sqrt{3}$. Removing the free set, of probability $1/16$, costs at most $A/4$ by Cauchy–Schwarz. Therefore, with $\eta = 1/\sqrt{3} - 1/4 > 1/4$ and $\beta = \varepsilon B$,

$$\mathcal{D}(\bar{F}) - L_0 \geq \eta A - \beta. \tag{24}$$

Let $E' = \varepsilon V - \sqrt{c}\, L_0$. Because $Q(F) \geq \sqrt{c}\sum_s |\widehat{\bar{F}}(s)| \geq \sqrt{c}\,\mathcal{D}(\bar{F})$, equations (22)–(24), and also the flat lower bound itself, imply

$$E' \geq \kappa\beta, \qquad E' \geq \sqrt{c}\,\eta A - \sqrt{c}\,\beta + \kappa\beta.$$

Take a convex combination with weight $\kappa/[\sqrt{c}(1+\eta)]$ on the second inequality. This gives

$$E' \geq \frac{\kappa\eta}{1+\eta}(A + \beta) \geq \frac{\kappa}{5}(A + \beta). \tag{25}$$

On a fixed (χ_C, z) fiber, (12) and the affine Lipschitz bound give $|F(x) - F(x')| \leq (A+\beta)\|x - x'\|_2$. Each such fiber is connected by moves flipping two coordinates within one block, with the core also regarded as a block. If signs vary on a fiber, a sign-changing edge has output difference at least one and length $2\sqrt{2}$, so $A + \beta \geq 1/(2\sqrt{2})$.

Otherwise the sign is constant on every fiber. Failure of (20) means that at some logical control value both core parity classes have the same sign. Fix physical controls with that value. The margin gives $|\mathbb{E}_{X_C} F| \geq 1/2$. Meanwhile (13) and the affine core-mean bound give $|\mathbb{E}_{X_C} F| \leq \sqrt{D+1}(A+\beta)$. In either case, $A + \beta \geq 1/(2\sqrt{D+1})$. Equation (25) and an infimum over representations prove the claim. □

Proposition 7 (Reduction after calibration)**.** *After calibration, and after any subsequent labels for which at least one star remains consistent, every minimum-variation sign completion has the form* (20)*. Every non-code sign completion has strictly larger variation cost than every star.*

Proof. It is convenient to compare costs multiplied by ε. A non-code sign completion has scaled cost at least

$$B_{\mathrm{nc}} := \sqrt{c}\, L_0 + \frac{\kappa}{10\sqrt{D+1}}. \tag{26}$$

By (16), every star has scaled cost at most

$$\sqrt{c + D}\, L_0 \leq \sqrt{c}\, L_0 + \frac{3D}{4\sqrt{c}} < B_{\mathrm{nc}}, \tag{27}$$

using $c = 10^8(D+1)^3 + 1$ and $\kappa = 1/200$. Later labels retain the calibration lower bound and a remaining star upper witness. If a sign completion is odd and code-invariant, its projection has the same signs and global margin, and (16) gives a code score of no larger cost. The remaining scaled optimization is therefore the finite weighted Fourier margin problem. □

4 The strictly ordered latent completions

Write $w(s) = a_s$ and

$$B_w(f) = \sum_s w(s)|\widehat{f}(s)|, \qquad \widehat{w}(v) = \frac{1}{M}\sum_s w(s)(-1)^{s\cdot v}.$$

The weights satisfy $w(s) \geq w(0) = \sqrt{c}$.

Lemma 8 (Signs and distinctness of the weight transform)**.** *Every nonconstant Walsh coefficient $\widehat{w}(v)$ is strictly negative, and all M values $\widehat{w}(v)$ are distinct.*

Proof. The integral representation

$$\sqrt{x} = \frac{1}{2\sqrt{\pi}}\int_0^\infty (1 - e^{-xt})t^{-3/2}\,dt$$

gives, for $v \neq 0$,

$$\widehat{w}(v) = -\frac{1}{2\sqrt{\pi}M}\int_0^\infty e^{-ct}\prod_{i:v_i=1}(1-e^{-r_i t})\prod_{i:v_i=0}(1+e^{-r_i t})\,t^{-3/2}\,dt < 0.$$

The integral is convergent at zero because $v \neq 0$.

For distinctness, all distinct integers in $[c, c+D]$ have different squarefree parts. Indeed, if $n = qu^2 < n' = qv^2$ have the same squarefree part, then $n' - n \geq q(2u+1) > 2\sqrt{c} > D$, a contradiction. Square roots of distinct squarefree integers are rationally linearly independent. Consequently the distinct values $\sqrt{n}$, $c \leq n \leq c+D$, are rationally linearly independent.

The coefficients of $M\widehat{w}(v)$ as a linear combination of these square roots are the coefficients of

$$P_v(t) = \prod_{i:v_i=1}(1-t^{r_i})\prod_{i:v_i=0}(1+t^{r_i}).$$

The odd integers q_i lie strictly between $2m$ and $4m$, so none divides another. The cyclotomic polynomial Φ_{q_i} divides $1 - t^{2q_j}$ exactly when $i = j$, and never divides $1 + t^{2q_j}$: a root of odd order cannot have a power equal to -1. Thus Φ_{q_i} divides P_v exactly when $v_i = 1$. All the polynomials P_v are distinct, proving the claim. □

Lemma 9 (Subgroup classification costs)**.** *If a proper subgroup $J \leq G$ has codimension at least two, the unique minimum-B_w real score with margin $1/2$ and positive set exactly J is $f_J = \mathbf{1}_J - 1/2$. In particular, writing C_0 for the singleton cost and C_v for the cost of $\{0, v\}$,*

$$C_0 = \frac{1}{2}\sqrt{c} + \widehat{w}(0) - \frac{2\sqrt{c}}{M}, \tag{28}$$

$$C_v = \frac{1}{2}\sqrt{c} + \widehat{w}(0) + \widehat{w}(v) - \frac{4\sqrt{c}}{M}. \tag{29}$$

All C_v are distinct and each is strictly smaller than C_0. These are latent weighted costs; the corresponding neural variation costs are C_0/ε and C_v/ε.

Proof. A norm-dual function has Fourier multipliers $+w(s)$ on $J^\perp \setminus \{0\}$, $-w(0)$ at zero, and zero elsewhere. In physical coordinates it is

$$\lambda_J(x) = \sum_{s \in J^\perp} w(s)(-1)^{s\cdot x} - 2w(0).$$

On J it is at least $(|J^\perp| - 2)w(0) > 0$. For $x \notin J$, Fourier inversion gives

$$\sum_{s \in J^\perp} w(s)(-1)^{s\cdot x} = \frac{M}{|J|} \sum_{h \in J} \widehat{w}(x+h) < 0,$$

so $\lambda_J(x) < 0$. Its normalized pairing with a feasible score is therefore minimized exactly when all signed margins equal $1/2$. That score also saturates the Fourier dual inequality, proving optimality and uniqueness. Direct evaluation gives (28)–(29). The final claims follow from Lemma 8, since $C_0 - C_v = 2\sqrt{c}/M - \widehat{w}(v) > 0$. □

Lemma 10 (Non-subgroup gap)**.** *Suppose a globally margin-$1/2$ score f has a positive set A containing zero, and let $\delta = |A|/M < 1/3$. If A is not a subgroup, then*

$$\sum_s |\widehat{f}(s)| \geq \frac{2-\delta}{1+2\delta}.$$

For every calibration-consistent positive set this lower bound is at least $61/38$.

Proof. Put $B = \sum_s |\widehat{f}(s)|$ and $a = \widehat{f}(0)$. Since $|f(x)| \leq B$, $a \leq \delta B - (1-\delta)/2$. Choose $u, v \in A$ with $u + v \notin A$. Every nonconstant character takes the value $+2$ or -2 in $\chi(0) + \chi(u) + \chi(v) - \chi(u+v)$, while the constant character gives $+2$. Hence

$$2 \leq f(0) + f(u) + f(v) - f(u+v) \leq 2a + 2(B - |a|).$$

If $a < 0$, combining these inequalities gives $(1+2\delta)B \geq 2 - \delta$. If $a \geq 0$, the mean bound gives $B \geq (1-\delta)/(2\delta)$, which is at least the claimed bound because $\delta < 1/3$. Calibration allows positives only in $\{0\} \cup U$, so $\delta \leq 1/16 + 1/M \leq 3/32$, giving $61/38$. □

Thus every non-subgroup code completion has scaled variation cost $\varepsilon\|F\|_{\mathcal{V}}$ at least

$$B_{\mathrm{ns}} := \frac{61}{38}\sqrt{c} > \frac{3}{2}\sqrt{c+D}, \tag{30}$$

which is larger than every star cost. A subgroup contained in $\{0\} \cup U$ has at most two elements: two distinct elements of U would sum to a nonzero element of H, which is forbidden. The only possible subgroup positive sets are therefore the singleton and the stars. These statements remain true when some points of U have subsequently been labeled negative.

5 A fixed sequence and strict variation gaps

Order the points of U uniquely by

$$C_{v_1} < C_{v_2} < \cdots < C_{v_K}, \qquad K = |U| = 2^{m-4}.$$

First present every calibration input in any fixed order. Next, for $j = 1, \ldots, K$, present one fixed physical representative x_j with $z(x_j) = v_j$ and $\chi_C(x_j) = +1$. These inputs were absent from the

calibration, are distinct, and all have true label -1. Complete the sequence to a permutation of the remaining cube points in any fixed order.

Before round j, the star at v_j is consistent with every preceding label and predicts $+1$ at x_j. Any odd code completion consistent with the history has positive set contained in $\{0\} \cup \{v_j, \dots, v_K\}$. By the preceding results, among all globally margin-valid completions the cheapest sign function is uniquely the star at v_j. More quantitatively, there is $\Delta_j > 0$ such that every history-consistent score predicting the correct label at x_j satisfies

$$\|F\|_{\mathcal{V}} \geq \frac{C_{v_j} + \Delta_j}{\varepsilon}. \tag{31}$$

Explicitly, one may take

$$\Delta_j = \min(\{B_{\mathrm{nc}}, B_{\mathrm{ns}}, C_0\} \cup \{C_{v_\ell} : j < \ell \leq K\}) - C_{v_j} > 0.$$

No quantitative lower bound on these algebraic differences is needed.

Only the initial calibration uses complete fibers. Later individual queries eliminate the corresponding code stars. The initial flat lower bound continues to hold, and every remaining star remains a global upper witness, so Proposition 7 still excludes all non-code signs at every designated round.

6 Every sufficiently large finite width and the posterior

Lemma 11 (Canonical energy and integer allocation)**.** *For every width and every network,*

$$K_N(W, v) \geq 2\sqrt{d} \sum_a |v_a| \|W_a\|_2 \geq 2\sqrt{d}\, \|F_{W,v}\|_{\mathcal{V}}. \tag{32}$$

For every fixed star score $\chi_C f_v$, there are networks at all sufficiently large integer widths whose outputs equal that score and whose energies converge to $2\sqrt{d}\, C_v/\varepsilon$.

Proof. The lower bound is the arithmetic–geometric mean inequality applied to each neuron's two energy terms. Apply Lemma 4 to every nonzero coefficient b_s in the star's code expansion. Index the resulting oriented atoms by $\ell = (s, j)$, $1 \leq j \leq 4$. Their incoming norms are $a_\ell = a_s$ and their signed readouts t_ℓ have absolute value $|b_s|/(4\varepsilon)$. Consequently

$$\sum_\ell |t_\ell| = L_0/\varepsilon, \qquad \sum_\ell |t_\ell| a_\ell = C_v/\varepsilon.$$

Define

$$p_\ell = \frac{|t_\ell|}{\sqrt{d}\, a_\ell}, \qquad P_* := \sum_\ell p_\ell \leq \frac{3}{2\varepsilon\sqrt{dc}} < 1.$$

At width N, take $n_\ell = \lceil N p_\ell \rceil$ replicas of each oriented atom, with readout t_ℓ/n_ℓ, and set all other neurons to zero. For all sufficiently large N, $\sum_\ell n_\ell \leq N$. Each orientation retains its exact total readout t_ℓ, even when the replica counts differ, so the affine cancellation and the whole score are unchanged. The energy is

$$\sum_\ell \left[d \frac{n_\ell}{N} a_\ell^2 + \frac{t_\ell^2}{n_\ell/N} \right] \longrightarrow 2\sqrt{d} \sum_\ell |t_\ell| a_\ell = \frac{2\sqrt{d}}{\varepsilon} C_v.$$

This convergence is along all sufficiently large integer widths, not only a subsequence. □

There are finitely many designated rounds. Equations (31) and (32), together with Lemma 11, therefore give $N_0(m)$ such that for every $N \geq N_0(m)$ and every round j there is a wrong-label star witness with energy strictly smaller than the energy infimum of all correct-label completions. Slightly multiplying its readout weights by a factor greater than one creates strict global margins and preserves this strict energy gap. Enlarge $N_0(m)$ to at least $8m-4$; the padded target teacher then keeps every posterior along the full permutation well-defined.

Lemma 12 (Finite-dimensional Gaussian separation)**.** *Fix a width N and a history. Suppose the correct-label event within the strict global-margin posterior has energy at least b, while a wrong-label, history-consistent strict-margin parameter vector has energy strictly below b. Then the posterior correct-label probability tends to zero as $\sigma \downarrow 0$.*

Proof. Continuity and finiteness of the input cube give an open parameter ball around the wrong witness contained in the denominator event. Shrink it so its energy is at most $a < b$. In whitened coordinates the Gaussian density is proportional to $\sigma^{-p}\exp(-K_N/(2\sigma^2))$, with fixed finite $p = N(d+2)$. The ball probability is at least a positive constant times $\sigma^{-p}e^{-a/(2\sigma^2)}$.

Choose $\eta \in (0,1)$ with $(1-\eta)b > a$. A Gaussian exponential moment bounds the raw probability of $K_N \geq b$ by $\eta^{-p/2}e^{-(1-\eta)b/(2\sigma^2)}$. The ratio tends to zero. The normalizing constant for the pre-data conditioning on $\mathcal{M}$ cancels in this ratio. The argument controls the entire posterior event and makes no choice among tied parameter minima. □

Apply Lemma 12 at each designated history to obtain (7). A union bound over the finitely many rounds gives the posterior-sampling claim, without any independence requirement on the prediction draws. Taking the minimum of the finitely many small-variance thresholds gives the majority and 2/3 conclusions.

7 Teacher complexity and interpretation

The target has the explicit score

$$T_m(x) = \chi_C(x)\left[\sum_{i=1}^{m}\chi_{B_i}(x) - (m-1)\right]. \tag{33}$$

The bracket is $+1$ when every block parity is positive, and is at most -1 otherwise. Thus its sign is (6) and its absolute output is at least one. Expand (33) as the m characters $\chi_{C\cup B_i}$ and $m-1$ copies of $-\chi_C$. Each support has odd size. Lemma 4 represents each character by four ψ neurons with incoming weights zero or ± 1, zero biases, and readouts $\pm 1/(4\varepsilon) = \pm 3/4$. This is a width-$4(2m-1) = 8m-4$ unit-bounded teacher.

A Boolean circuit computes each block parity and the core parity, tests whether all block parities are positive, and combines with the core parity. With fan-in two this uses $O(d)$ gates. Every input is relevant, so any fan-in-two circuit has at least $d-1$ gates; the Boolean-circuit complexity is $\Theta(d)$. The displayed teacher has $O(md) = O(m^7)$ scalar parameters, all bounded by one. Hence every usual bounded same-depth teacher-size convention is polynomial in m, while the mistake count is 2^{m-4}.

Because $D = 6m^2$ and $c = 10^8(D+1)^3 + 1$, one has $d = \Theta(m^6)$. The lower bound is therefore stretched exponential in input dimension and Boolean-circuit size, and exponential in the explicit teacher width. The construction uses a simple but nonstandard activation with a periodic derivative. Strict monotonicity and a global Lipschitz bound are insufficient for a polynomial arbitrary-order mistake theorem; the corresponding questions for tanh, ReLU, GELU, and hardtanh remain distinct.

The proof uses the strong-prior limit at each fixed eligible width. It makes no assertion about a fixed positive variance, a width-uniform small-variance threshold, or the canonical width-first mean-field limit.

Remark 13 (Arbitrarily small periodic perturbations)**.** *The same argument works for every fixed activation $\psi(t) = \alpha t + \varepsilon\phi(t)$ with $\alpha > \varepsilon > 0$. Only the affine coefficient vector L changes, and all code costs are multiplied by $1/\varepsilon$. Enlarge the fixed core-size prefactor as a function of ε, if necessary, to keep the total ideal neuron allocation below one. If necessary, split each oriented readout into $\lceil 1/(4\varepsilon)\rceil$ copies to keep teacher weights bounded by one. In particular $t + \varepsilon\phi(t)$, for any fixed $0 < \varepsilon < 1$, is strictly increasing and uniformly within ε of the identity, and has the same qualitative lower bound. No uniformity as $\varepsilon \downarrow 0$ is asserted.*

A Proof of the atomic theorem

We give all quantitative ingredients of Theorem 3. The absolutely convergent Fourier series of the triangle wave is

$$\phi(t) = \sum_{n\geq 1 \text{ odd}} \alpha_n (-1)^{(n-1)/2} \sin(n\pi t/2), \qquad \alpha_n = \frac{8}{\pi^2 n^2}, \quad \sum_n \alpha_n = 1. \tag{34}$$

Adding an even integer to any incoming weight, including the bias, changes the ridge only by a global sign on the cube. Reduce each parameter modulo two into $[-1, 1]$, decreasing its norm and preserving all absolute Fourier coefficients. We first work with this reduced vector, of norm r. If $r = 0$, the triangle ridge is zero and the claims are immediate with $e_\theta = 0$. Hence assume $r > 0$ below.

A.1 Projection bounds and low norms

For a harmonic and an odd-size support A, its absolute Fourier coefficient is

$$|\cos(n\pi b/2)| \prod_{i\in A} |\sin(n\pi w_i/2)| \prod_{i\notin A} |\cos(n\pi w_i/2)|.$$

For block B_j, put $p_j = \prod_{i\in B_j} |\sin(n\pi w_i/2)|$ and $q_j = \prod_{i\in B_j} |\cos(n\pi w_i/2)|$. Hölder gives $p_j + q_j \leq 1$ for block size at least two. Consequently, with $\mathcal{L} = \sum_s |\widehat{h}_\theta(A_s)|$, one has

$$\mathcal{L} \leq 1, \qquad \mathcal{L} \leq 1 - \alpha_1(1 - P), \tag{35}$$

where P is the first-harmonic product

$$P = |\cos(\pi b/2)| \prod_{i\in C} |\sin(\pi w_i/2)| \prod_j (p_j + q_j).$$

For $0 \leq u \leq 1$, the triangle series gives

$$\sum_n \alpha_n \sin^2(n\pi u/2) = u, \qquad \sum_n \alpha_n \sin^2(n\pi u/2)\cos^2(n\pi u/2) = \tfrac{1}{2}\min(u, 1-u).$$

It follows that

$$\sum_n \alpha_n \sin^4(n\pi u/2) = u - \tfrac{1}{2}\min(u, 1-u) \leq \tfrac{1}{2}(u + u^2).$$

Hölder on the c core factors and $c \geq 4$ therefore imply, with $t = r/\sqrt{c}$,

$$\mathcal{L} \leq \frac{1}{c} \sum_{i \in C} \sum_{n} \alpha_n |\sin(n\pi w_i/2)|^c \leq \frac{t + t^2}{2}.$$

Thus for $r \leq .9\sqrt{c}$,

$$Q(h_\theta) \leq .95\sqrt{1.01}\, r < .955r, \tag{36}$$

since $D/c \leq .01$.

A.2 Localization near a pure atom

Suppose $r > .9\sqrt{c}$ and $P > .86$. For a first-harmonic block write $a = \max(p_j, q_j)$ and $b' = \min(p_j, q_j)$. Because its size is at least four, generalized Hölder gives $\sqrt{a} + \sqrt{b'} \leq 1$. Since $a + b' > .86$, it follows that $a > .85$ and

$$b' \leq (1 - \sqrt{a})^2 \leq .273(1 - a)^2, \qquad -\log a \leq 1.05[-\log(a + b')].$$

For the last elementary inequality, put $u = 1 - a \leq .15$, use $\log(1 + .273u^2/a) \leq .273u^2/a$ and $-\log a \geq u + u^2/2$.

Choose the nearest integer vector u: bias even, core coordinates odd, and every coordinate of a block odd when its sine product dominates or even when its cosine product dominates. Reduced parameters and the strict localization imply that this is a minimal pure atom: its coordinates are zero or ± 1 as specified in the theorem. Put $\delta = \theta - u$ and $\rho = \|\delta\|_2$. The dominant products are products of $\cos(\pi\delta_i/2)$. Since $\cos(\pi s/2) \leq e^{-\pi^2 s^2/8}$ for $|s| \leq 1$,

$$\rho^2 \leq \frac{8}{\pi^2}\, 1.05(-\log P) < .1296, \qquad \rho < .36.$$

If ρ_j is the perturbation norm on one block, its nondominant product is at most $(\pi^4/256)\rho_j^4$. Every dominant product is at least .85. Therefore

$$\log P \leq -\frac{\pi^2}{8}\rho^2 + \frac{\pi^4}{256 \cdot .85}\rho^4 \leq -1.17\rho^2.$$

For $\rho \leq .36$ this implies $1 - P \geq \rho^2$. Since $\alpha_1 > .8$, (35) yields

$$\mathcal{L} \leq 1 - .8\rho^2. \tag{37}$$

Let the pure ridge be $\eta\chi_{A_T}$, let $a = \sqrt{|A_T|}$, and put $a_{\max} = \sqrt{c + D}$ and $\Delta = a_{\max} - a \leq D/(2\sqrt{c}) \leq .05$. The 1-Lipschitz property gives $|\widehat{h}_\theta(A_T)| \geq 1 - \mathbb{E}|\delta \cdot \bar{X}| \geq 1 - \rho$. Hence

$$Q(h_\theta) \leq a\mathcal{L} + \Delta(\mathcal{L} - |\widehat{h}_\theta(A_T)|) \leq a - .8a\rho^2 + \Delta\rho.$$

Using $r \geq a - \rho$, we obtain

$$r - Q(h_\theta) \geq .8a\rho^2 - (1 + \Delta)\rho. \tag{38}$$

A.3 A strict projection gap for an absolute linear form

For an affine linear form $Z = \delta_0 + \sum_i \delta_i X_i$ and $\rho = \|\delta\|_2$, we claim

$$\sum_{\emptyset \neq S \subseteq [m]} \left| \widehat{|Z|}\left(\bigcup_{j \in S} B_j \right) \right| \leq \mathbb{E}|Z| - \gamma\rho, \qquad \gamma = \frac{2}{\pi}\left(1 - e^{-1/2} - \frac{1}{16}\right) > .21. \tag{39}$$

Use $|z| = (2/\pi)\int_0^\infty (1-\cos(tz))t^{-2}\,dt$. At frequency t, put $p_j = \prod_{i\in B_j} |\sin(t\delta_i)|$ and $q_j = \prod_{i\in B_j} |\cos(t\delta_i)|$. The absolute sum of nonempty block-character coefficients of the cosine is bounded by

$$N_t = r_t \left[\prod_j (p_j + q_j) - \prod_j q_j \right],$$

where r_t is the absolute product of the bias cosine and cosine factors outside the blocks. Since $p_j + q_j \leq 1$,

$$N_t \leq 1 - |\cos(t\delta_0) \prod_i \cos(t\delta_i)| \leq D_t := 1 - \cos(t\delta_0) \prod_i \cos(t\delta_i).$$

For $t\rho \leq 1$, all cosine factors are positive. The inequality $\cos s \leq e^{-s^2/2}$ gives $D_t \geq (1 - e^{-1/2})t^2\rho^2$. By AM–GM and block sizes at least four,

$$N_t \leq \sum_j p_j \leq t^4\rho^4/16 \leq t^2\rho^2/16.$$

Integrating the nonnegative difference $D_t - N_t$ over $0 \leq t \leq 1/\rho$ proves (39).

A.4 Completing the norm and quantitative estimates

If $\rho \leq 1.5/\sqrt{c}$, then $\max_x |\delta \cdot \bar{x}| \leq 1.5\sqrt{(c + D + 1)/c} < 1.51 < 2$. At an odd lattice peak the exact triangle identity is therefore

$$h_\theta(x) = \eta\chi_{A_T}(x)(1 - |\delta \cdot \bar{x}|).$$

Multiplication by χ_{A_T} permutes the block characters, and (39) gives

$$Q(h_\theta) \leq a[1 - \mathbb{E}|Z|] + a_{\max}[\mathbb{E}|Z| - \gamma\rho] \leq a - (\gamma a_{\max} - \Delta)\rho.$$

Consequently,

$$r - Q(h_\theta) \geq (\gamma a_{\max} - \Delta - 1)\rho \geq .15\rho. \tag{40}$$

If instead $\rho \geq 1.5/\sqrt{c}$, then (38) gives the same lower bound $.15\rho$.

If $r > .9\sqrt{c}$ but $P \leq .86$, then

$$\frac{r - Q(h_\theta)}{r} \geq 1 - \frac{[1 - .14(8/\pi^2)]\sqrt{1.01}}{.9} > .010.$$

Together with (36), this proves the atomic inequality and gives strictness except at pure atoms. The pure atoms attain equality directly, establishing (10) and all equality cases. Undoing the modulo-two reduction can only increase the norm.

For quantitative rounding, set $e_\theta = r$ in the two far cases and $e_\theta = \rho$ in the localized case. The preceding estimates imply (11) with $\kappa = 1/200$. In a far case the atom is r-Lipschitz, and at fixed physical controls its affine argument has square mean at most $(D+1)r^2$ over the core. This proves (12)–(13) there.

In a localized case, periodicity gives the global identity

$$\phi((u+\delta)\cdot\bar{x}) = \eta\chi_{A_T}(x)\zeta(\delta\cdot\bar{x}), \qquad \zeta(t) = \phi(1+t).$$

The function ζ is even and 1-Lipschitz. On a fixed (χ_C, z) fiber the character is constant, proving (12). Pairing either value of any one core coordinate in the core average gives

$$|\mathbb{E}_{X_C}\phi((u+\delta)\cdot\bar{X})| \le \min_{i\in C}|\delta_i| \le \rho/\sqrt{c} \le \sqrt{D+1}\,e_\theta,$$

which proves (13).

Finally, in a far case $\|h_\theta\|_2 \le r = e_\theta$, so orthogonal projection gives (14). In a localized case let $g = \eta\chi_{A_T}$ be the rounded pure ridge. Its code projection is itself, and the Lipschitz property gives $\|h_\theta - g\|_2 \le \|\delta\cdot\bar{X}\|_2 = \rho$. Consequently $\|(I-P)h_\theta\|_2 \le \|h_\theta - g\|_2 \le e_\theta$. Global output signs from reduction modulo two do not change any of these estimates. This completes the theorem.

References

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